

Tutorial 10: PCA and Factor Analysis

ENVX2001 – Applied Statistical Methods

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Semester 1

PCA and Factor Analysis

This tutorial uses football-player performance data from Wilson et al. (2017), *Skill not athleticism predicts individual variation in match performance of soccer players*. The study measured semi-professional football players from the University of Queensland Football Club and asked whether morphology, balance, athleticism and motor skill predicted performance in soccer-tennis and 11-a-side matches.



In the paper, the authors measured:

- **Balance:** average time before losing balance while standing on high-density foam with eyes closed.
- **Athletic performance:** 1500 m speed, wall-squat endurance, jump distance, 40 m sprint speed and agility-course speed.
- **Motor skill:** dribbling speed, juggling ability, volley accuracy, passing accuracy and heading accuracy.

The key biological question is whether complex match performance is better explained by general athletic capacity or football-specific motor skill.

We are going to use principal components analysis and factor analysis to interpret the multivariate structure of these player performance traits.

Downloads

File	Used in	Download
football_player_performance.csv	PCA and factor analysis	Download
football_goals_20_games_24_players.csv	Goals prediction	Download

Save both files into a folder called data inside your project folder. The code in this tutorial expects to find them at data/football_player_performance.csv and data/football_goals_20_games_24_players.csv.

Setup

Load these packages to make the plots look better:

```
CODE
library(EFAtools)
library(ggplot2)
library(ggfortify)
```

Load the data

First open the player performance dataset. The CSV file is saved in the data folder.

Later in the tutorial, we will also use a second independent dataset, football_goals_20_games_24_players.csv, to practise prediction using regression.

```
CODE
players <- read.csv("data/football_player_performance.csv", header = TRUE)
```

Check the variable names:

CODE

```
names(players)
```

OUTPUT

```
[1] "player" "balance" "X1500m" "squat" "jump" "sprint" "agility"
[8] "drib" "jugg" "voll" "pass" "head"
```

Because R changes column names that begin with a number, 1500m may appear as X1500m.

CODE

```
head(players)
```

OUTPUT

```
  player balance  X1500m squat jump  sprint agility  drib  jugg
1      1   6.485 4.297994   361 2.67 7.272727 3.601286 2.545455 14.80000
2      2  25.000 4.602194   100 2.55 7.214200 3.530339 2.554162 26.50000
3      3   3.920 3.916449    87 2.51 6.430868 3.412034 2.471042 10.11111
4      4  20.000 3.605769    85 2.36 6.700168 3.227666 2.175814 14.50000
5      5  46.700 4.065041   141 2.52 6.920415 3.446154 2.471042  8.40000
6      6  60.000 3.778338   172 2.53 6.666667 3.530339 2.621416 21.15385
      voll  pass  head
1 11.76471 20.40816 15.15152
2 11.00000 31.40816 14.49275
3 11.76471 17.85714 14.49275
4 12.80855 18.87122 13.70566
5 10.73989 21.81725 16.55008
6 11.62791 24.39024 15.38462
```

CODE

```
str(players)
```

OUTPUT

```
'data.frame': 24 obs. of 12 variables:
 $ player : int  1 2 3 4 5 6 7 8 9 10 ...
 $ balance: num  6.49 25 3.92 20 46.7 ...
 $ X1500m : num  4.3 4.6 3.92 3.61 4.07 ...
 $ squat : int  361 100 87 85 141 172 80 216 81 189 ...
 $ jump : num  2.67 2.55 2.51 2.36 2.52 2.53 2.19 2.38 2.42 2.72 ...
 $ sprint: num  7.27 7.21 6.43 6.7 6.92 ...
 $ agility: num  3.6 3.53 3.41 3.23 3.45 ...
 $ drib : num  2.55 2.55 2.47 2.18 2.47 ...
 $ jugg : num  14.8 26.5 10.1 14.5 8.4 ...
 $ voll : num  11.8 11 11.8 12.8 10.7 ...
 $ pass : num  20.4 31.4 17.9 18.9 21.8 ...
 $ head : num  15.2 14.5 14.5 13.7 16.6 ...
```

Description of the variables

CODE

```
summary(players)
```

OUTPUT

```
  player      balance      X1500m      squat
Min.   : 1.00   Min.   : 3.92   Min.   :3.606   Min.   : 79.00
```

1st Qu.: 6.75	1st Qu.:12.00	1st Qu.:3.940	1st Qu.: 84.75
Median :12.50	Median :22.92	Median :4.364	Median :103.00
Mean :12.50	Mean :25.48	Mean :4.283	Mean :131.50
3rd Qu.:18.25	3rd Qu.:34.99	3rd Qu.:4.549	3rd Qu.:166.00
Max. :24.00	Max. :60.00	Max. :4.854	Max. :361.00
jump	sprint	agility	drib
Min. :2.050	Min. :6.231	Min. :3.228	Min. :2.176
1st Qu.:2.360	1st Qu.:6.650	1st Qu.:3.379	1st Qu.:2.504
Median :2.465	Median :6.932	Median :3.497	Median :2.548
Mean :2.439	Mean :6.889	Mean :3.466	Mean :2.553
3rd Qu.:2.535	3rd Qu.:7.175	3rd Qu.:3.545	3rd Qu.:2.632
Max. :2.720	Max. :7.491	Max. :3.666	Max. :2.784
jugg	voll	pass	head
Min. : 7.842	Min. : 7.804	Min. :10.81	Min. : 4.182
1st Qu.:10.659	1st Qu.: 9.501	1st Qu.:20.41	1st Qu.:14.296
Median :14.082	Median :11.250	Median :28.67	Median :15.268
Mean :15.900	Mean :11.017	Mean :29.77	Mean :14.973
3rd Qu.:18.990	3rd Qu.:12.544	3rd Qu.:38.00	3rd Qu.:17.712
Max. :39.857	Max. :13.889	Max. :50.00	Max. :19.231

The variables are:

- balance: static balance score
- X1500m: speed over 1500 m
- squat: wall-squat endurance time
- jump: standing jump distance
- sprint: 40 m sprint speed
- agility: speed through an agility course
- drib: dribbling speed through an agility course
- jugg: juggling or keep-up ability
- voll: volley-kick accuracy
- pass: passing accuracy
- head: heading accuracy

Higher values generally indicate better performance on that trait.

Correlation matrix

Let's first do a correlation matrix. We exclude the player ID column and analyse the performance traits only.

CODE

```
cor(players[, 2:12])
```

OUTPUT

	balance	X1500m	squat	jump	sprint
balance	1.0000000000	-0.120666590	0.056071952	0.005165273	0.14975877
X1500m	-0.120666590	1.000000000	0.159548821	0.096566292	0.34226606
squat	0.056071952	0.159548821	1.000000000	0.432157704	0.30883252
jump	0.005165273	0.096566292	0.432157704	1.000000000	0.33871992
sprint	0.149758766	0.342266059	0.308832520	0.338719924	1.00000000

```

agility -0.229848417 0.168169479 0.445175401 0.571266808 0.16286933
drib    0.470845996 0.356356256 0.159653733 0.154769848 0.29294354
jugg    0.187921370 0.155839614 0.004928929 0.280997027 -0.07650198
voll    0.397624644 0.006100075 0.153281966 0.025642532 0.32241412
pass    0.416714230 0.155833542 -0.042274047 -0.006327859 0.25026128
head    -0.006255931 0.032500939 0.122024732 0.081488984 -0.10767999
      agility      drib      jugg      voll      pass
balance -0.22984842 0.47084600 0.187921370 0.397624644 0.416714230
X1500m  0.16816948 0.35635626 0.155839614 0.006100075 0.155833542
squat   0.44517540 0.15965373 0.004928929 0.153281966 -0.042274047
jump    0.57126681 0.15476985 0.280997027 0.025642532 -0.006327859
sprint  0.16286933 0.29294354 -0.076501981 0.322414125 0.250261285
agility  1.00000000 0.02135601 0.185489117 -0.303537885 -0.015220367
drib     0.02135601 1.00000000 0.346830908 0.313925571 0.507806703
jugg     0.18548912 0.34683091 1.000000000 0.107128520 0.112221815
voll     -0.30353789 0.31392557 0.107128520 1.000000000 0.530011484
pass     -0.01522037 0.50780670 0.112221815 0.530011484 1.000000000
head     0.36781797 0.07426933 0.235577420 0.394861742 0.341212829
      head
balance -0.006255931
X1500m  0.032500939
squat   0.122024732
jump    0.081488984
sprint  -0.107679989
agility  0.367817975
drib     0.074269329
jugg     0.235577420
voll     0.394861742
pass     0.341212829
head     1.000000000

```

CODE

```
Corrmatrix <- cor(players[, 2:12])
```

A correlation matrix lets us see which traits tend to vary together. For example, if several skill traits are positively correlated, this suggests that some players have generally high technical ability across multiple football-specific tasks.

Bartlett's Test of Sphericity

Now we can do Bartlett's Test of Sphericity. This test compares the correlation matrix to an identity matrix. If it is significant, it is worth doing a PCA.

CODE

```
BARTLETT(Corrmatrix, N = nrow(players), cor_method = c("pearson"))
```

OUTPUT

```
✓ The Bartlett's test of sphericity was significant at an alpha level of .05.
  These data are probably suitable for factor analysis.
```

```
 $\chi^2(55) = 81.19, p = 0.012$ 
```

A significant result means the variables are sufficiently correlated for a dimension-reduction method such as PCA or factor analysis.

Principal Components Analysis

Note: to make this a PCA based on a correlation matrix, we have to scale the variables, hence `scale = TRUE`. There are two main principal components functions, but they are very similar. Note `prcomp` calls the loadings “rotations”, not to be confused with rotations below.

CODE

```
pca1 <- prcomp(players[, 2:12], scale = TRUE)
pca2 <- princomp(players[, 2:12], cor = TRUE)
```

CODE

```
summary(pca1)
```

OUTPUT

Importance of components:

	PC1	PC2	PC3	PC4	PC5	PC6	PC7
Standard deviation	1.6999	1.4767	1.1795	1.0726	1.02453	0.80985	0.79892
Proportion of Variance	0.2627	0.1982	0.1265	0.1046	0.09542	0.05962	0.05803
Cumulative Proportion	0.2627	0.4609	0.5874	0.6920	0.78742	0.84704	0.90507

	PC8	PC9	PC10	PC11
Standard deviation	0.60575	0.55566	0.52948	0.29702
Proportion of Variance	0.03336	0.02807	0.02549	0.00802
Cumulative Proportion	0.93842	0.96649	0.99198	1.00000

CODE

```
summary(pca2)
```

OUTPUT

Importance of components:

	Comp.1	Comp.2	Comp.3	Comp.4	Comp.5
Standard deviation	1.6999027	1.4766764	1.1794865	1.0726210	1.02452713
Proportion of Variance	0.2626972	0.1982339	0.1264717	0.1045924	0.09542326
Cumulative Proportion	0.2626972	0.4609311	0.5874028	0.6919951	0.78741840

	Comp.6	Comp.7	Comp.8	Comp.9	Comp.10
Standard deviation	0.80984729	0.7989246	0.6057459	0.55566144	0.52948396
Proportion of Variance	0.05962297	0.0580255	0.0333571	0.02806906	0.02548666
Cumulative Proportion	0.84704137	0.9050669	0.9384240	0.96649302	0.99197968

	Comp.11
Standard deviation	0.297024364
Proportion of Variance	0.008020316
Cumulative Proportion	1.000000000

The first principal component describes the major axis of variation among players. In the Wilson et al. paper, separate PCAs were used to summarise athletic performance and motor skill. Here, we are doing one combined PCA across balance, athletic performance and motor skill traits, so the first few components may separate general performance, athleticism and football-specific skill.

Loadings

Let's look at the loadings. Called "rotations" in `prcomp` and "loadings" in `princomp`. They are the Pearson's correlation between that variable and that Principal Component.

CODE

```
pca1
```

OUTPUT

```
Standard deviations (1, .., p=11):
[1] 1.6999027 1.4766764 1.1794865 1.0726210 1.0245271 0.8098473 0.7989246
[8] 0.6057459 0.5556614 0.5294840 0.2970244

Rotation (n x k) = (11 x 11):
      PC1      PC2      PC3      PC4      PC5      PC6
balance 0.2936386 -0.35087385 -0.05458821 -0.01236935 -0.50256454 0.22417684
X1500m  0.2299058 0.15552197 -0.28197657 -0.42123014 0.58080664 -0.18787938
squat   0.2547779 0.35550940 -0.14404714 0.34331707 -0.16954618 -0.19781100
jump    0.2656109 0.43501720 -0.01783357 0.04579993 -0.35433245 -0.03857511
sprint  0.3261362 0.10208432 -0.52353137 0.20056752 0.10706161 -0.05512777
agility 0.1673372 0.55847807 0.21211288 0.02473830 0.03452894 0.42295776
drib    0.4327319 -0.14166176 -0.12392103 -0.36532953 -0.05741543 0.19918699
jugg    0.2447494 0.06217972 0.37349976 -0.55564980 -0.25869776 -0.42767336
voll    0.3619836 -0.32312886 0.07293446 0.39291316 0.07224943 -0.49119815
pass    0.3945243 -0.28979483 0.10341401 0.06052617 0.23371625 0.47674183
head    0.2392728 0.04926387 0.63756754 0.24769351 0.33626830 -0.02753477

      PC7      PC8      PC9      PC10     PC11
balance 0.20538168 -0.54118171 0.365326098 -0.12428219 -0.02713210
X1500m  0.18718969 -0.15409295 0.479562718 -0.01307712 -0.06663040
squat   0.65671346 0.09650932 -0.081047782 0.33932493 0.19978295
jump    -0.42236751 0.37267671 0.478977025 -0.18468479 0.17304917
sprint  -0.42420682 -0.39669448 -0.404284981 -0.05135179 0.22582954
agility -0.01039040 -0.22522259 -0.142378623 0.02285589 -0.60231545
drib    0.25091726 0.42491622 -0.384734195 -0.45697622 -0.02280376
jugg    -0.15523189 -0.17263458 -0.237654632 0.35303879 0.05601670
voll    -0.10513326 0.15078192 0.048968103 -0.04463885 -0.56584548
pass    -0.17259657 0.22907881 0.099702647 0.59253599 0.14422004
head    0.07178534 -0.20375286 -0.003474513 -0.38133775 0.40809767
```

CODE

```
loadings(pca2)
```

OUTPUT

```
Loadings:
      Comp.1 Comp.2 Comp.3 Comp.4 Comp.5 Comp.6 Comp.7 Comp.8 Comp.9 Comp.10
balance 0.294 0.351          0.503 0.224 0.205 0.541 0.365 0.124
X1500m  0.230 -0.156 0.282 0.421 -0.581 -0.188 0.187 0.154 0.480
squat   0.255 -0.356 0.144 -0.343 0.170 -0.198 0.657          -0.339
jump    0.266 -0.435          0.354          -0.422 -0.373 0.479 0.185
sprint  0.326 -0.102 0.524 -0.201 -0.107          -0.424 0.397 -0.404
agility 0.167 -0.558 -0.212          0.423          0.225 -0.142
drib    0.433 0.142 0.124 0.365          0.199 0.251 -0.425 -0.385 0.457
jugg    0.245          -0.373 0.556 0.259 -0.428 -0.155 0.173 -0.238 -0.353
voll    0.362 0.323          -0.393          -0.491 -0.105 -0.151
pass    0.395 0.290 -0.103          -0.234 0.477 -0.173 -0.229          -0.593
head    0.239          -0.638 -0.248 -0.336          0.204          0.381

      Comp.11
balance
X1500m
```

```
squat -0.200
jump -0.173
sprint -0.226
agility 0.602
drib
jugg
voll 0.566
pass -0.144
head -0.408
```

	Comp.1	Comp.2	Comp.3	Comp.4	Comp.5	Comp.6	Comp.7	Comp.8	Comp.9
SS loadings	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
Proportion Var	0.091	0.091	0.091	0.091	0.091	0.091	0.091	0.091	0.091
Cumulative Var	0.091	0.182	0.273	0.364	0.455	0.545	0.636	0.727	0.818

	Comp.10	Comp.11
SS loadings	1.000	1.000
Proportion Var	0.091	0.091
Cumulative Var	0.909	1.000

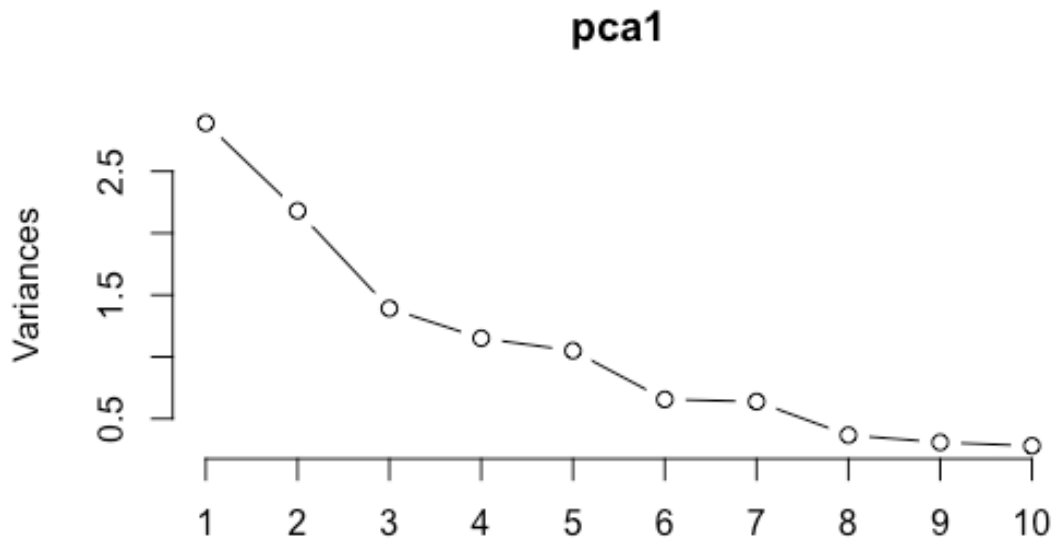
Interpretation:

- Variables with large positive or negative loadings contribute strongly to that component.
- If all traits load in the same direction on PC1, PC1 can be interpreted as general player performance.
- If athletic traits load in one direction and skill traits load in another, the component may represent a contrast between athletic capacity and football-specific skill.

Screeplot

To do a screeplot, follow the commands below. Note this is the standard deviations, which are just square root of the variances or eigenvalues.

```
CODE
screeplot(pca1, type = "lines")
```

The screeplot helps decide how many components are worth interpreting. A common rule of thumb is to keep components before the curve flattens out.

Principal Component Scores

To get your principal components scores for plotting and analysis, do the following:

CODE

```
pca1$x
```

OUTPUT

	PC1	PC2	PC3	PC4	PC5	PC6
[1,]	1.22265294	3.21272139	-0.84513278	1.729875647	-0.41800665	-1.16536979
[2,]	1.01549076	0.72909627	-0.07487739	-1.126299782	0.08030236	-0.44330634
[3,]	-1.95431230	0.13616424	0.62019420	0.683133943	-0.08692596	-0.65692178
[4,]	-2.74171773	-1.49202883	0.50124852	1.735454370	-0.78052160	-1.85418850
[5,]	-0.29218316	-0.08747326	-0.11988125	1.108831773	-0.99391905	0.32842292
[6,]	0.91997987	-0.28734416	0.84641464	0.232906297	-2.57193000	0.29232622
[7,]	-0.95858539	-2.60131264	1.43635426	0.990763299	0.22009132	0.24168604
[8,]	0.04244428	0.86220012	-1.13368412	0.678632536	-0.23885615	0.30250751
[9,]	-1.12244287	-0.06955280	1.12603291	0.288229193	-0.12664793	1.91010162
[10,]	3.39686967	0.83136647	-0.43475555	1.121845842	0.67921147	0.65544911
[11,]	1.10678403	-2.25184331	-2.80892930	-1.372109965	-1.76088086	0.26168616
[12,]	1.62463419	-0.31440598	-0.62813600	-0.915492585	-0.10688046	-1.21576231
[13,]	-0.57673686	1.50371749	1.09522239	0.357669243	-0.40536591	1.07877742
[14,]	-2.45940301	1.55096778	-1.73104596	-0.815402453	0.44465656	-0.23066431
[15,]	-1.83385997	-1.38084282	-0.43730279	-0.678128599	0.95811502	0.19042700
[16,]	0.42353253	-0.66991754	0.33476898	1.159555934	1.82618886	0.02417214
[17,]	2.33632926	-2.16227916	-0.18130741	-0.003407567	0.72452040	0.03044801

[18,]	-1.18106445	1.01946486	-1.42004479	0.469515124	-0.72656471	0.99789886
[19,]	-0.28781540	1.77196130	0.32939063	-1.408868336	0.66689434	-0.33260253
[20,]	0.77348049	0.82720824	3.03423799	-2.239773715	-1.30000966	-0.51164628
[21,]	2.10946365	-1.29318452	0.20187351	0.034487363	0.57647043	0.44954616
[22,]	-3.17285972	-0.60941350	-0.65733911	-1.356394009	0.79164937	0.24990229
[23,]	0.16649597	1.91778410	0.60759087	-0.489210312	1.41139229	0.25681745
[24,]	1.44282322	-1.14305374	0.33910755	-0.185813240	1.13701651	-0.85970707
	PC7	PC8	PC9	PC10	PC11	
[1,]	1.06626147	0.74992918	-0.609228418	0.46483136	0.17305201	
[2,]	-1.09718405	-0.55422064	0.020706681	0.30612703	-0.11318983	
[3,]	-0.48184786	1.35102821	0.176215524	-0.67519862	-0.43378351	
[4,]	-1.06403599	-0.28180079	0.378695770	0.62881031	0.21732729	
[5,]	0.19619194	-0.83106823	0.854657258	-0.89586615	0.36499683	
[6,]	0.70036296	-0.63152387	0.184658562	-0.46021429	-0.40732572	
[7,]	0.47708118	0.29020781	-0.199234399	-0.10259401	0.08118763	
[8,]	1.15124813	-0.95643173	-0.276070905	-0.34757651	0.10353943	
[9,]	-0.72084050	0.16806957	0.038709675	1.07069258	0.30262699	
[10,]	-0.65278915	0.16655620	0.932371171	0.05236096	-0.10214435	
[11,]	-0.12849364	0.55873899	0.297675862	0.89339944	-0.12124663	
[12,]	0.78971314	0.03142220	-0.228192952	0.24684205	0.39712028	
[13,]	0.84348491	0.35975295	-0.505277549	0.58132253	-0.50547940	
[14,]	-0.33609876	-0.32114157	0.801087448	-0.11434646	-0.54458835	
[15,]	0.95117438	-0.59904114	-0.831463657	0.01884134	-0.16940143	
[16,]	0.03885635	-1.02979447	-0.006515691	0.66914191	-0.25696656	
[17,]	-0.25250279	-0.04466779	-0.360784257	-0.27940159	0.17394913	
[18,]	-1.68675889	0.24054300	-1.225329831	-0.62610765	0.31313624	
[19,]	-0.98192696	-0.61152963	-0.431957026	-0.35399954	0.06783876	
[20,]	-0.05633572	-0.03278367	-0.008561725	0.05457287	0.13508982	
[21,]	0.02130256	0.74598418	0.172521694	-0.78114525	-0.03614367	
[22,]	1.20492210	0.59954232	0.449245111	-0.24242268	0.33615101	
[23,]	0.36131351	0.36550291	0.878867693	0.12037610	0.36393275	
[24,]	-0.34309832	0.26672601	-0.502796039	-0.22844571	-0.33967872	

So to combine them with the original variables, do this:

```
CODE
playerpcscores <- cbind(players, pca1$x)
```

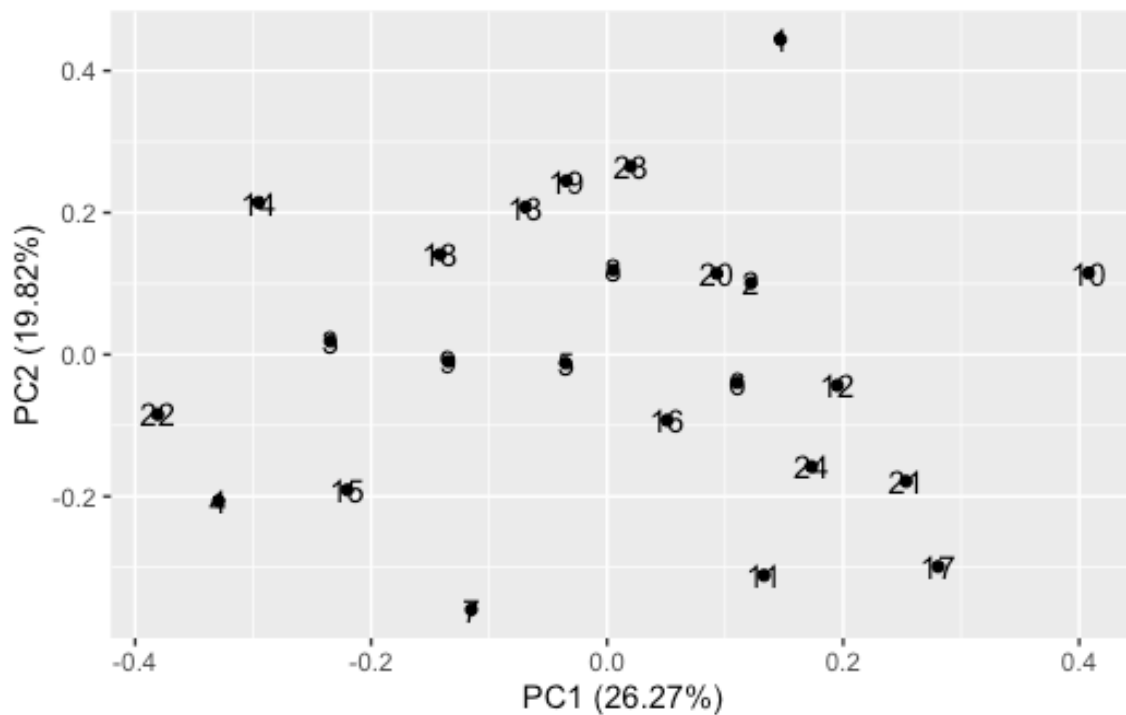
Each player now has a score on each principal component. These scores tell us where each player sits on the multivariate axes of football performance.

Biplot

We can produce a biplot in ggplot2. Here the player number is used as a label.

```
CODE
autoplot(pca1, data = playerpcscores, label = TRUE, label.label = "player")
```

```
OUTPUT
Warning: `aes_string()` was deprecated in ggplot2 3.0.0.
i Please use tidy evaluation idioms with `aes()`.
i See also `vignette("ggplot2-in-packages")` for more information.
i The deprecated feature was likely used in the ggfortify package.
Please report the issue at <https://github.com/sinhrks/ggfortify/issues>.
```



We could add the loadings, but they can be a bit messy. It is often best to look at the loadings or rotations tables directly.

You can make the plot using different symbols, change background etc through reading the ggplot2 documentation.

Regression: Goals vs PCA

We now use a simple regression to link the PCA results to an outcome variable.

We will relate **goals scored over 20 games** to the first principal component (PC1), which represents overall performance.

Read in the goals dataset:

```
CODE
goals <- read.csv("data/football_goals_20_games_24_players.csv", header = TRUE)
```

For simplicity, we will match players by row order and combine the goals dataset with the PCA scores. We avoid using the object name data, because data() is also a built-in R function.

```
CODE
pca_scores <- as.data.frame(pca1$x)
```

```
# Keep only PC1 for this regression
pca_scores <- pca_scores[, "PC1", drop = FALSE]

# Check that the PCA data and goals data have the same number of players
nrow(pca_scores)
```

OUTPUT

```
[1] 24
```

CODE

```
nrow(goals)
```

OUTPUT

```
[1] 24
```

CODE

```
combined <- cbind(
  pca_scores,
  goals
)
```

Simple regression: Goals vs PC1

CODE

```
LM_goals_PC1 <- lm(goals_20_games ~ PC1, data = combined)
summary(LM_goals_PC1)
```

OUTPUT

```
Call:
lm(formula = goals_20_games ~ PC1, data = combined)

Residuals:
    Min       1Q   Median       3Q      Max
-2.6108 -0.9792  0.1929  0.8037  3.4207

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  7.58333    0.33493   22.642  <2e-16 ***
PC1          -0.09529    0.20127   -0.473    0.641
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.641 on 22 degrees of freedom
Multiple R-squared:  0.01009,    Adjusted R-squared:  -0.03491
F-statistic: 0.2241 on 1 and 22 DF,  p-value: 0.6406
```

Interpretation:

- The response variable is goals_20_games
- The predictor is PC1
- A positive relationship means players with higher overall performance (PC1) score more goals

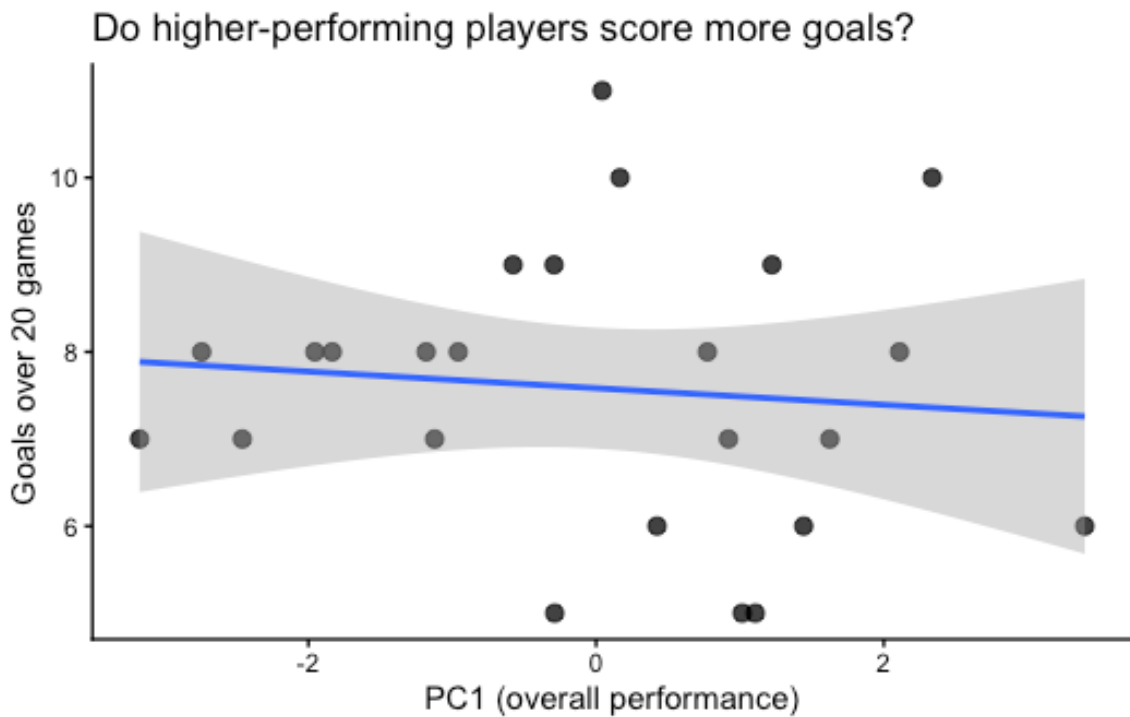
Scatterplot

CODE

```
ggplot(combined, aes(x = PC1, y = goals_20_games)) +  
  geom_point(size = 2.5, alpha = 0.8) +  
  geom_smooth(method = "lm", se = TRUE) +  
  theme_classic() +  
  labs(  
    x = "PC1 (overall performance)",  
    y = "Goals over 20 games",  
    title = "Do higher-performing players score more goals?"  
  )
```

OUTPUT

```
`geom_smooth()` using formula = 'y ~ x'
```



Interpretation:

- Each point is a player
- Points close together have similar performance and goal output
- The slope shows whether better overall players tend to score more goals

Key point

This links PCA to prediction:

- PCA identifies the main performance axis (PC1)
- Regression tests whether that axis predicts goals

Factor Analysis with Varimax Rotation

To do a Varimax rotation on your principal components, follow these commands. Do a factor analysis with the rotation being Varimax.

Here we start with three factors because the paper describes three broad biological domains: balance, athletic performance and motor skill.

CODE

```
fa1 <- factanal(players[, 2:12], 3, rotation = "varimax")
fa1
```

OUTPUT

Call:

```
factanal(x = players[, 2:12], factors = 3, rotation = "varimax")
```

Uniquenesses:

balance	X1500m	squat	jump	sprint	agility	drib	jugg	voll	pass
0.722	0.920	0.656	0.524	0.504	0.005	0.706	0.913	0.171	0.607
head									
0.005									

Loadings:

	Factor1	Factor2	Factor3
balance	0.524		
X1500m	0.105	0.261	
squat	0.112	0.567	0.102
jump		0.687	
sprint	0.441	0.525	-0.158
agility	-0.444	0.801	0.395
drib	0.468	0.273	
jugg		0.168	0.226
voll	0.849		0.322
pass	0.538	0.133	0.292
head			0.993

	Factor1	Factor2	Factor3
SS loadings	1.935	1.905	1.427
Proportion Var	0.176	0.173	0.130
Cumulative Var	0.176	0.349	0.479

Test of the hypothesis that 3 factors are sufficient.
The chi square statistic is 19.84 on 25 degrees of freedom.
The p-value is 0.755

The rotated factor loadings may help identify clusters of variables. For example:

- one factor may represent athletic performance
- one factor may represent football-specific motor skill
- one factor may represent balance or a narrower technical component

Try changing the number of factors and compare whether the interpretation becomes clearer or less clear:

CODE

```
fa2 <- factanal(players[, 2:12], 2, rotation = "varimax")
fa2
```

OUTPUT

Call:
factanal(x = players[, 2:12], factors = 2, rotation = "varimax")

Uniquenesses:

balance	X1500m	squat	jump	sprint	agility	drib	jugg	voll	pass
0.669	0.932	0.747	0.633	0.805	0.005	0.629	0.894	0.359	0.497
head									
0.712									

Loadings:

	Factor1	Factor2
balance	0.573	
X1500m	0.133	0.224
squat		0.497
jump		0.606
sprint	0.336	0.286
agility	-0.320	0.945
drib	0.570	0.216
jugg	0.194	0.262
voll	0.799	
pass	0.677	0.212
head	0.253	0.473

	Factor1	Factor2
SS loadings	2.09	2.027
Proportion Var	0.19	0.184
Cumulative Var	0.19	0.374

Test of the hypothesis that 2 factors are sufficient.
The chi square statistic is 33.83 on 34 degrees of freedom.
The p-value is 0.476

CODE

```
fa4 <- factanal(players[, 2:12], 4, rotation = "varimax")
fa4
```

OUTPUT

Call:
factanal(x = players[, 2:12], factors = 4, rotation = "varimax")

Uniquenesses:

balance	X1500m	squat	jump	sprint	agility	drib	jugg	voll	pass
0.677	0.839	0.618	0.511	0.521	0.005	0.005	0.815	0.005	0.555
head									
0.101									

Loadings:

	Factor1	Factor2	Factor3	Factor4
balance	0.423		0.367	
X1500m		0.167	0.364	
squat		0.602		
jump		0.686	0.117	
sprint	0.338	0.541	0.150	-0.220
agility	-0.485	0.747	0.116	0.434
drib	0.291		0.950	
jugg			0.353	0.229

voll	0.966		0.235	
pass	0.465	0.394	0.264	
head	0.176		0.929	

	Factor1	Factor2	Factor3	Factor4
SS loadings	1.804	1.742	1.510	1.293
Proportion Var	0.164	0.158	0.137	0.118
Cumulative Var	0.164	0.322	0.460	0.577

Test of the hypothesis that 4 factors are sufficient.
The chi square statistic is 9.89 on 17 degrees of freedom.
The p-value is 0.908

Final interpretation

Wilson et al. found that motor skill was a stronger predictor of soccer-tennis and 11-a-side match performance than general athleticism. In this tutorial, PCA and factor analysis are used to explore the same biological idea from the trait data: whether football players vary mostly along a general performance axis, or whether athletic and technical skill traits form separate multivariate dimensions.

The independent goals dataset then extends the tutorial into prediction. Instead of asking only how traits group together, the regression asks whether a player's score on the main PCA axis predicts goals scored over 20 games.